

Puzzle 02 — Google's Hat Puzzle

ITAI-1370 · History of Artificial Intelligence · Houston Community College

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Task 1 — The Main Puzzle

Puzzle source: puzzles.nigelcoldwell.co.uk/thirtynine.htm

One hundred prisoners are each assigned a randomly colored hat — either red or blue. Each prisoner can see all the hats in front of them but cannot see their own. Starting from the back of the line, each prisoner must call out a single color. If they call their own hat color correctly, they survive. The prisoners may agree on a strategy beforehand but may not communicate once the game begins.

The Parity Strategy

The optimal solution relies on **parity** — whether a count is odd or even. Before the game, all prisoners agree on one rule: Person 1 counts all red hats visible in front of them. If the count is odd, they call **red**; if even, they call **blue**. This single call broadcasts the starting parity to the entire group. Every person from position 2 onward tracks the reds already called, counts what they see ahead, and deduces their own hat color with 100% accuracy. Only Person 1 faces a genuine 50/50 chance — making them the willing sacrifice that guarantees the survival of the other 99.

The interactive learning tools below were built to visualize this process step by step:

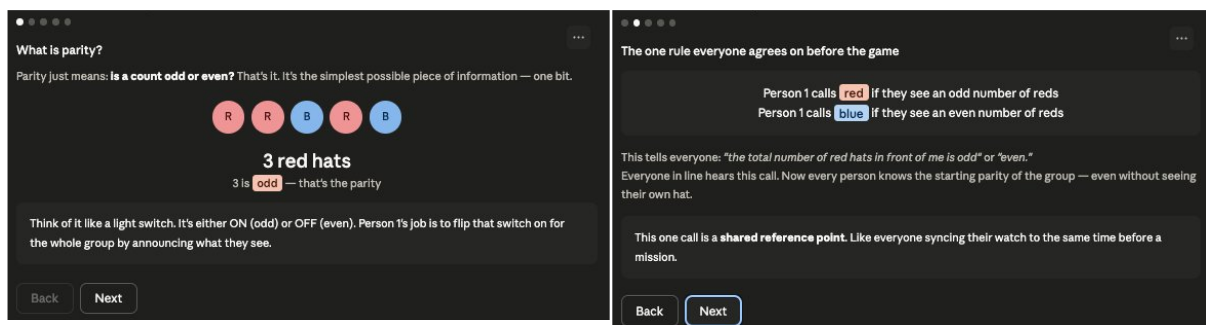


Figure 1 (left): Parity defined — is a count odd or even? One bit of information. Figure 2 (right): The single agreed-upon rule that makes the entire strategy work.

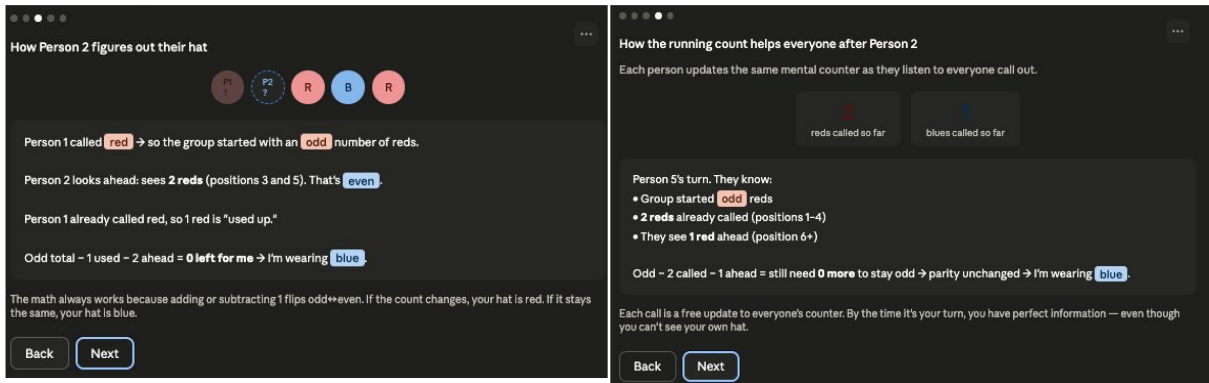


Figure 3 (left): How Person 2 deduces their own hat using Person 1's signal. Figure 4 (right): How every subsequent person uses the running count of reds called.

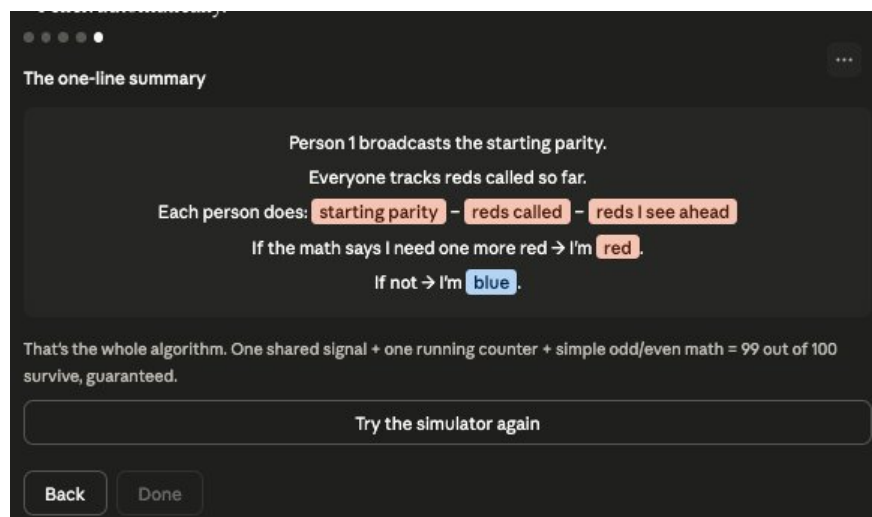


Figure 5: The one-line algorithm — starting parity minus reds called minus reds ahead = your hat color.

Interactive Simulator & Logic Tree

The following tools were used to step through the deduction person by person and map the full decision logic of the parity strategy:

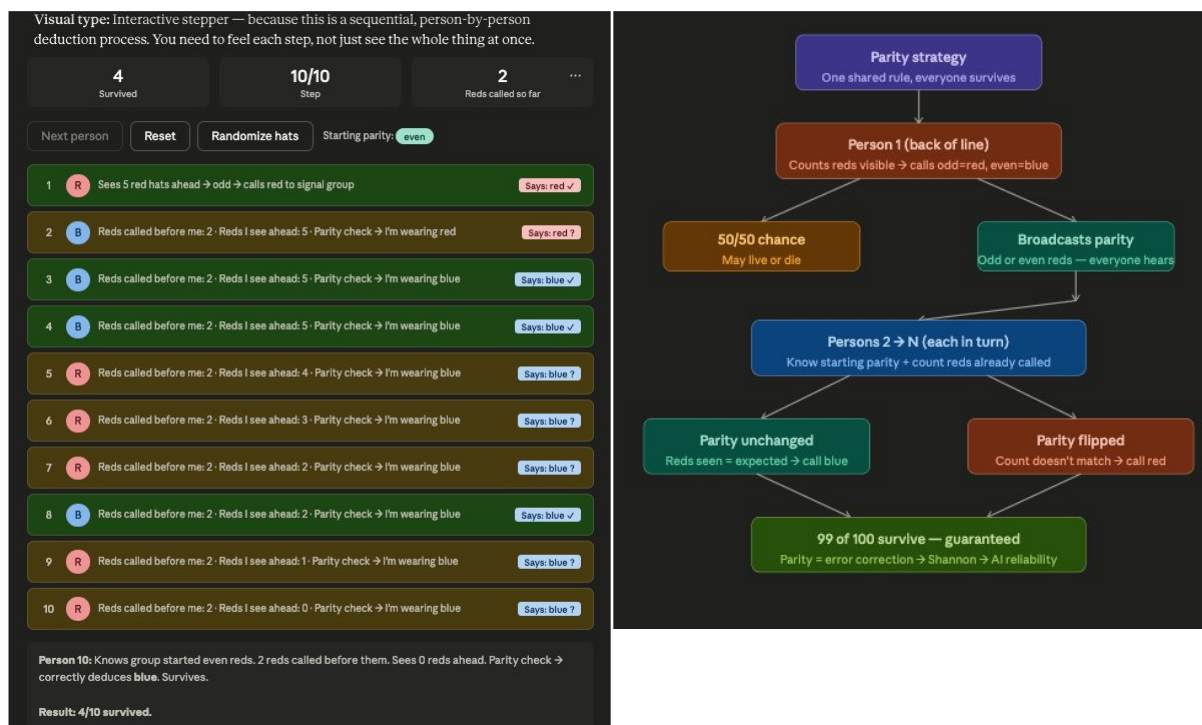


Figure 6 (left): Interactive 10-person simulator — each row shows the parity check and deduction in real time.

Figure 7 (right): Full logic tree mapping the strategy from Person 1's sacrifice through guaranteed group survival.

Task 2 — AI Problem Solving Connections

The hat puzzle is not merely a mathematical curiosity — it is a concrete model of the foundational problems that artificial intelligence is designed to solve: making optimal decisions under incomplete information using shared rules and inference. Each prisoner functions as an AI agent operating with limited environmental observation, yet arriving at the correct output through structured reasoning. The following connections illustrate how the puzzle maps directly onto core AI concepts.

Parity Bits and Error Correction

Claude Shannon's 1948 paper *A Mathematical Theory of Communication* introduced the concept of encoding a single bit of information to detect and correct errors in data transmission. Person 1's call in the hat puzzle is precisely that parity bit — and every subsequent person uses it to error-correct their own guess. This mechanism is embedded in every digital device in use today.

Binary Classification

Every person in the puzzle performs a binary classification task: red or blue. AI classifiers — logistic regression models, neural networks, support vector machines — do exactly this. Given a set of observable features, they output one of two categories with maximum accuracy. The hat puzzle is a human-scale demonstration of that process.

Bayesian Inference

Each prisoner continuously updates their belief about their own hat color as new evidence arrives — each call heard updates the mental model. This is Bayesian reasoning: prior belief plus new evidence equals updated posterior. AI systems use this framework in natural language processing, medical diagnosis, spam filtering, and recommendation systems.

Multi-Agent Coordination

The prisoners form a cooperative multi-agent system in which no single agent has complete information, yet the group achieves near-perfect collective outcomes through a shared protocol. Modern AI uses multi-agent reinforcement learning frameworks with shared reward functions to solve exactly these kinds of coordination problems — from autonomous vehicle fleets to distributed robotics.

Spin-Off: More Colours and Multi-Class Classification

The extension of the puzzle to three or more hat colors maps directly to multi-class classification in AI. Instead of a single parity bit, multiple agents at the back each encode one additional bit of information — equivalent to adding output nodes to a neural network classifier. Scaling the sacrifice scales the precision of the group's inference.

Task 3 — Reflective Narrative

When I first encountered this puzzle, my immediate reaction was that it was unsolvable. The premise felt impossible: you cannot know what you cannot see. I read the instructions more than once and kept arriving at the same conclusion — there is no way a prisoner could correctly identify their own hat color without more information. That assumption turned out to be the exact obstacle I needed to overcome.

The breakthrough came when I stopped trying to solve it individually and started looking at the group as a system. The puzzle is not asking any single person to have complete information. It is asking the group to share one carefully chosen piece of information — and from that single signal, for everyone else to derive what they need. Once I understood that Person 1 does not need to save themselves, that their role is to give everyone else what is needed to save themselves, everything shifted. That is a form of sacrifice — one agent absorbing uncertainty so that 99 others can operate with certainty.

To genuinely understand the mechanics, I worked through the puzzle interactively using custom-built visual tools — the parity explainer, the step-by-step deduction simulator, and the full logic tree diagram shown in this document. It took several passes through the material before the logic fully locked in. The concept of parity — simply whether a count is odd or even — felt abstract at first. But once I understood it as a light switch, either on or off, the entire strategy became clear. One bit of shared information was sufficient to guarantee the survival of 99 out of 100.

What struck me most about this puzzle is that none of it would work without a pre-agreed framework. The parity rule had to be established before the game began. Without that shared protocol, each prisoner would be guessing randomly at 50/50 odds. The rule itself is the intelligence — not any individual prisoner. This connects directly to how AI systems function. A model does not operate on complete information about the world. It operates on learned patterns, shared frameworks, and inference. The hat puzzle is a simplified but structurally accurate model of that process.

The connection to Claude Shannon's information theory was particularly meaningful to me. Shannon proved in 1948 that a single parity bit could detect errors in data transmission — that one carefully chosen piece of information could protect an entire stream of data. The hat puzzle is a physical, human-scale demonstration of that same principle. And it is embedded in every device I use daily — in hard drives, network protocols, and memory chips. Understanding that made the puzzle feel less like an abstract riddle and more like a window into the foundational logic of the digital world.

This exercise also connected to something more personal. There have been times in my own life where I refused to move or make a decision until I felt I had it completely figured out. That kind of thinking — waiting for complete information before acting — is almost never viable in real

circumstances. What I have learned, both through this puzzle and through lived experience, is that optimal decisions do not require complete information. They require the right shared framework and the discipline to apply it consistently. Sometimes I am given enough information to infer, to gather collective input, and to position myself within a structure that gets me to the next step. With persistence and consistency, the outcome can be close to — or exactly — what I was aiming for.

The biggest lesson this puzzle taught me is that intelligence — whether human or artificial — is not about having all the answers. It is about having the right rules, applying them with discipline, and trusting that the framework will produce the outcome. That was faith-inspiring to engage with, and it is a principle I intend to carry forward.

References: Shannon, C. E. (1948). A mathematical theory of communication. Bell System Technical Journal, 27, 379-423. | Puzzle source: puzzles.nigelcoldwell.co.uk/thirtynine.htm | Interactive diagrams created with Claude (Anthropic) visual learning tools, June 2026.